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Generative design of main landing gear for a remote-controlled aircraft

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ABSTRACT

Current research and development in the aviation and aerospace industry aim to increase efficiency by reducing the weight of a remote-controlled (RC) aircraft using various simulation tools. Advancements in weight reduction algorithms have led to the development of sophisticated topology optimization and generative design algorithms. In the present paper, the main landing gear of an RC aircraft has been designed using a fusion of topology optimization and generative design algorithms. A finite element analysis (FEA) has been performed using ANSYS to obtain total deformation and von-Mises stress for various designs of the landing gear. The ratio for a factor of safety to mass has been calculated for each design of the main landing gear obtained through conventional design, topology optimized design, generative design, and fusion of topology optimization and generative design. The trade-off for each design has been obtained to reduce the overall weight of the RC aircraft. The final design of the main landing gear resulted in a 30% weight reduction from the conventional design.
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1. Introduction

Unmanned aerial vehicles (UAVs) are a generic class of aerial vehicles that can function without an onboard human pilot. Their use began as target drones and missiles in the first World War [1]. However, it wasn't until the 1960s that UAVs were used for more sophisticated purposes, such as carrying important payloads and military reconnaissance [2]. The development of autonomous technology has further boosted the growth of the UAV market [3]. Since the second world war, there have been rapid advances in the aeronautical and avionics industries. The world has witnessed the use of small UAVs [4,5] in fields such as agriculture, military, photography, mapping, and even reconnaissance because of the reliability, reusability, accessibility, and better maneuverability of UAVs. The fixed-wing and vertical take-off or landing (VTOL) are the two popular classifications of UAVs that are employed to carry and transport heavy payloads and for longer distances. Over the years fixed-wing UAVs have proven to be lighter, more efficient, economical, and better suited for missions that require high payload fractions when compared to a VTOL [6]. Much research has been carried out to further increase the weight-carrying capacity of a fixed-wing UAV. One such research aimed at reducing the overall weight of a fixed-wing UAV by using 3D-printed carbon fiber reinforced thermoplastics on topology optimized structures [7]. The design and build of the UAV must be such that its components weigh as little as possible without compromising the structural requirements for safety and reusability. Several components including landing gear undergoes severe stress, every time a UAV takes off or lands. The main landing gear is an extremely important component and the safety of the UAV relies on its structural soundness of it.

In this paper, topology optimization and generative design have been used to expand the scope of weight reduction in a fixed-wing UAV by exploring new landing gear designs. Topology optimization is a sophisticated structural design method that can produce the optimal structural configuration through reasonable material distribution while meeting specified load conditions, performance, and constraints [8]. On the other hand, generative design is a tool that optimizes a design using mathematical formulations and algorithms to minimize or maximize an objective function [9]. Both are complex yet effective structural optimization techniques that have been put to the test in order to create a structurally strong and stiff design that is capable of replacing the conventional landing gear design with the advantage of being as lightweight as possible. Furthermore, the scope of this paper is expanded to include designing a landing gear with a fusion of topology optimization and generative design. The final design of the main landing gear has been cho-

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sen based on a factor of safety (FOS) by mass ratio and the results have been successfully validated using a static structural FEA using ANSYS. This paper aims to pave the way for generative design to revolutionize the aviation industry by developing lighter aircraft.

2. Force calculation for the main landing gear

The main landing gear of UAV experiences 80–85 % share of the total static load because of its location near the center of gravity [10]. The landing gear must withstand a vertical impact force equal to 3 times the weight of the aircraft during landing according to the American Institute of Aeronautics and Astronautics (AIAA). There is a high chance that landing will not be as conceptualized as the RC plane is controlled manually. The maximum vertical force acting on the landing gear is calculated using Eq. (1) [11].

$$Y\prime = 3gm \tag{1}$$

where, Y' is the maximum vertical force acting on the landing gear and is calculated as 117.72 N ≈ 120 N and m is the mass of the RC plane considered as 4 kg.

The frictional force $F_{z\prime}$ and change in momentum $F_{z^{\cdot}}$ in the horizontal direction is calculated using Eqs. (2) and (3) respectively [12].

$$F_{z\prime} = Z\prime\mu \tag{2}$$

$$F_{z^{\cdot}} = \frac{mv_z^2}{2s} \tag{3}$$

where Z' is the horizontal force opposite to the travel direction, μ is the friction coefficient = 0.3, v_z is the horizontal approach velocity = 10 m/s and s is the stopping distance = 10 m. The total horizontal force acting on the landing gear is calculated as 56 N ≈ 60 N. An equal distribution of forces between the two legs of the landing gear has been assumed, resulting in 30 N as horizontal force and 60 N as vertical force on each leg.

3. Design methodology for landing gear

In the present paper, various design methodologies have been discussed which include conventional, topology optimization, generative design, and fusion of generative with topology optimized design. The detailed design procedures including fabrication methodology based on selected materials have been discussed in this section.

3.1. Conventional design approach

The conventional design methodology is primarily based on manufacturing simplicity to enable production with less time and expense. The simplicity of production makes replacement simple when necessary. The conventional design includes material selection, fundamental concepts of machine design principles and methodology.

For the landing gear of a UAV, Al 6061T6 is chosen due to its lightweight and good strength. The chemical composition and material properties of Al 6061T6 are represented in Table 1 and Table 2 respectively. The design is to be calculated for bending, compression, buckling, and torsion under the calculated load. But before that, the angle of landing gear is to be decided such that the tail of the RC plane doesn't contact the ground at any point during take-off or landing. The 45° is most suitable for the landing gear considering the ground clearance and the wheelbase based upon the design of the RC plane. The landing gear is divided into different sections labeled 1-1′ and 2-2′ as shown in Fig. 1 for calculating the loads and designing a conventional main landing gear. Surface A represented in Fig. 1 is structurally constrained to be a fixed surface during the simulation, while the structural loads F_y and F_z will act on surface B.

Table 2
Mechanical properties of Al 6061T6.

Mechanical Properties	Metric
Ultimate Tensile Strength	310 MPa
Tensile Yield Strength	276 MPa
Shear Strength	207 MPa
Fatigue Strength	96.5 MPa
Modulus of Elasticity	68.9 GPa
Shear Modulus	26 GPa

The dimensions for landing gear are calculated based on critical bending stress conditions. The bending stress of about 1-1′ due to horizontal force is calculated using Eq. (4) [12].

$$S_{yt} = \sigma_b FOS \tag{4}$$

The yield strength S_{yt} for Al 6061 T6 material is considered 276 MPa with a FOS of 2.86. The working bending stress σ_b is calculated as 96.4 MPa which is used to calculate cross-section dimensions using Eq. (5) [12].

$$\sigma_b = \frac{Y\prime wy}{\frac{1}{12}bd^3} \tag{5}$$

where, Y' is the maximum vertical force acting on the landing gear, the distance w is taken as 129.19 mm, and by assuming $d = 2y$ and b as 40 mm, y is calculated as 1.74 mm ≈ 2 mm. So, the cross-section of the bar is 40 X 4 mm which is again checked under bending and torsion due to horizontal force.

The bending stress along 2-2′ is calculated based on horizontal force using Eq. (6) and from that factor of safety is obtained as 57 using Eq. (4). So, the design of landing gear under bending is quite safe under loading conditions.

$$\sigma_b = \frac{Z\prime hy}{\frac{1}{12}bd^3} \tag{6}$$

The design of landing gear is also checked under torsion conditions due to horizontal force and a factor of safety is obtained which is found to be safe. The dimensions of the conventional main landing gear design are depicted in Fig. 2 obtained through the calculation.

3.2. Topology optimization

Structural topology optimization has proven to be one of the most beneficial yet complex structural optimization techniques available to engineers and scientists [13]. Weight reduction in today's engineering industry plays a paramount role due to its direct effect on fuel efficiency and profit. The aviation industry has seen multiple instances [14–16] where engineers have employed the use of topology optimization as a tool to reduce

Table 1
Chemical composition of base metal Al 6061T6.

	Si	Mg	Cr	Cu	Al
wt. %	0.6	1.0	0.2	0.28	Balance

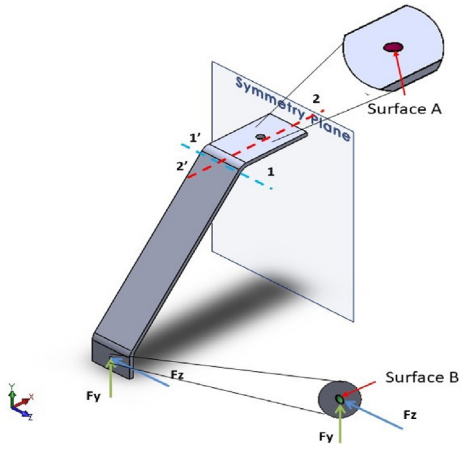


Fig. 1. Sectional view of a conventional main landing gear.

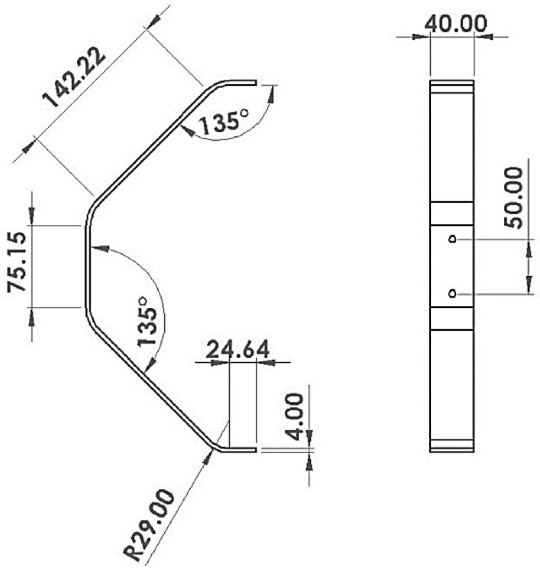


Fig. 2. Dimensions of conventional main landing gear.

the weight of a component under the input loads and constraints of the design domain. Multiple opportunities have been observed in the UAV industry [17] where weight could be reduced from a component without compromising its structural integrity. The topology optimization module of Autodesk Fusion 360 has been used on one such component of the UAV, which is the main landing gear. Once the predefined loads and constraints have been input into the conventional design, preserved regions have been selected around where the mass and volume needed to be preserved for easy handling and assembly. The material selected here is Al 6061 T6 and the preferred manufacturing method is considered to be CNC machining. At the end of the topology optimization result, the design acquired had a 40 % reduction in mass from the conventional design as shown in Fig. 3. The design on the left side of Fig. 3 represents the topology optimization setup of conventional main landing gear with 100 % mass. The topology optimization simulation is run for weight reduction of 40 % in Fusion 360 software through which the design is obtained on the right side of Fig. 3.

3.3. Generative design approach

The process of employing algorithms to investigate a design's variations beyond what is currently achievable using the standard design process is known as a generative design [18]. The generative design employs parameters and goals to quickly explore thousands of design variants to identify the optimal solution, mimicking nature's evolutionary process. The generative design approach tests the structure and learns from all of its iterations to develop a more ideal design that satisfies the objective function set by the user [19]. The advancements in aviation and UAV technology have seen the implementation of this evolutionary approach [20,21]. Autodesk's Fusion 360 provides access to generative design on its cloud-based server [22,23]. This feature has been used to initialize and compute the optimal design solutions for the main landing gear from the conventional design.

3.3.1. Geometry assignment

In order to create the generative design, various other additions must be made to apply boundary conditions and give bounds to the design which is to be generated. The starting shape acts as the original CAD model which is to be optimized using generative design. Preserved geometry is the geometry created to assign the boundary conditions such as fixed support and forces. Obstacle geometry and obstacle offset inform the artificial intelligence-based code to not generate designs that intersect with these areas. Once the geometry has been assigned, the expected loads and constraints have been applied to them according to the calculations done in Section 2. Fig. 4 displays the direction and magnitude of the loads applied.

3.3.2. Material selection

Material selection is an important aspect of designing any component undergoing loading conditions. Selection of the proper material and manufacturing method is to be the next major step in the design process. HP's multi-jet fusion [24] 3D printing method using nylon polyamide 12 (PA 12) has been thought of as the prime additive manufacturing method for a non-metallic

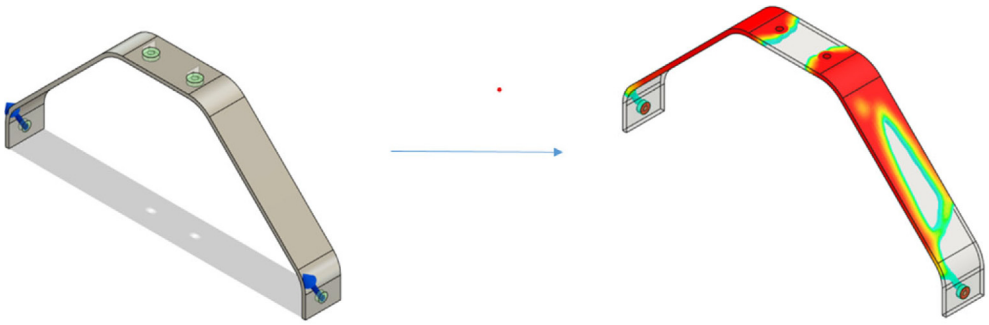


Fig. 3. Topology optimization of main landing gear with loading conditions [left side], and final structure obtained [right side].

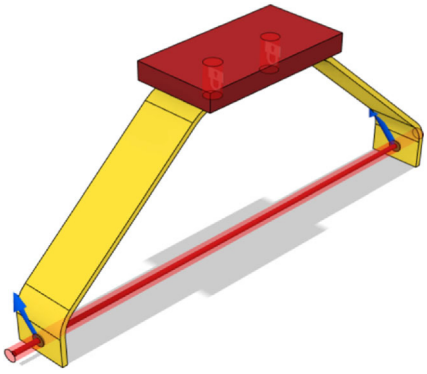


Fig. 4. Overview of the generative model.

Table 3
Material selection.

Material	Density (g/cc)	Fabrication Method
Orgasol Invent PA12	0.96	Material Jet Fusion (MJF)
Al 6061T6	2.7	5-axis CNC, Binder Jetting
AlSi10Mg	2.67	Direct Metal Laser Sintering (DMLS)

main landing gear design due to its strength, stiffness, and isotropic material properties [25]. AlSi10Mg, a lightweight and durable component popular among engineers for 3D printing, has been chosen [26]. Also, research [27,28] has proved the reliability and effectiveness of 3D-printed Al 6061T6 using the binder jetting method. Table 3 summarizes the material selection, respective densities, and appropriate fabrication technology required.

3.3.3. Objective function and model selection

The objective function and mass constraints have been specified, to generate landing gear designs using the artificial intelligence of the software. Stiffness has been chosen as the objective function to maximize. Fusion 360's generative design module produces a variety of designs satisfying the input criteria by the users, which in this case has been designed according to stiffness with a target mass of 160 gms and a factor of safety of 3. Fig. 5 displays a graph that provides information about the minimum factor of safety and mass provided by the design. Based on this plot, the design with the highest FOS/mass has been selected as the optimal design, as displayed in Fig. 5.

3.4. Fusion of topology optimization and generative design

The generative design approach gave us a design for the main landing gear which is quite complicated and expensive to manufacture. It would require a 5-axis CNC or binder jetting 3D printing technology to create such a complicated geometry from Al6061 T6. Topology optimization presented us with a design that is cheaper to manufacture. A 3-axis CNC could easily create the design on an already bent aluminium landing gear. Even though topology optimization shows more promise over generative design due to its ease in manufacturing, it must be noted that a significant fraction of the FOS/mass ratio is traded off. To get the best out of both designs, a completely new design has been made that combined the features of both topology optimization and generative design as shown in Fig. 6, which had a FOS/mass ratio greater than that of topology optimization and is easy to manufacture.

4. Finite element structural analysis

The landing gear undergoes various forces during touchdown, of which the main force is the normal force generated due to descent velocity, and the other is a horizontal force opposite to the motion of the plane, which tends to deflect the landing gear in a backward direction. The purpose of the structural analysis is to gather static loading information and ensure the integrity and dependability of the structure. Static structural analysis is performed in the predetermined loading and boundary conditions using ANSYS Workbench, a computational tool.

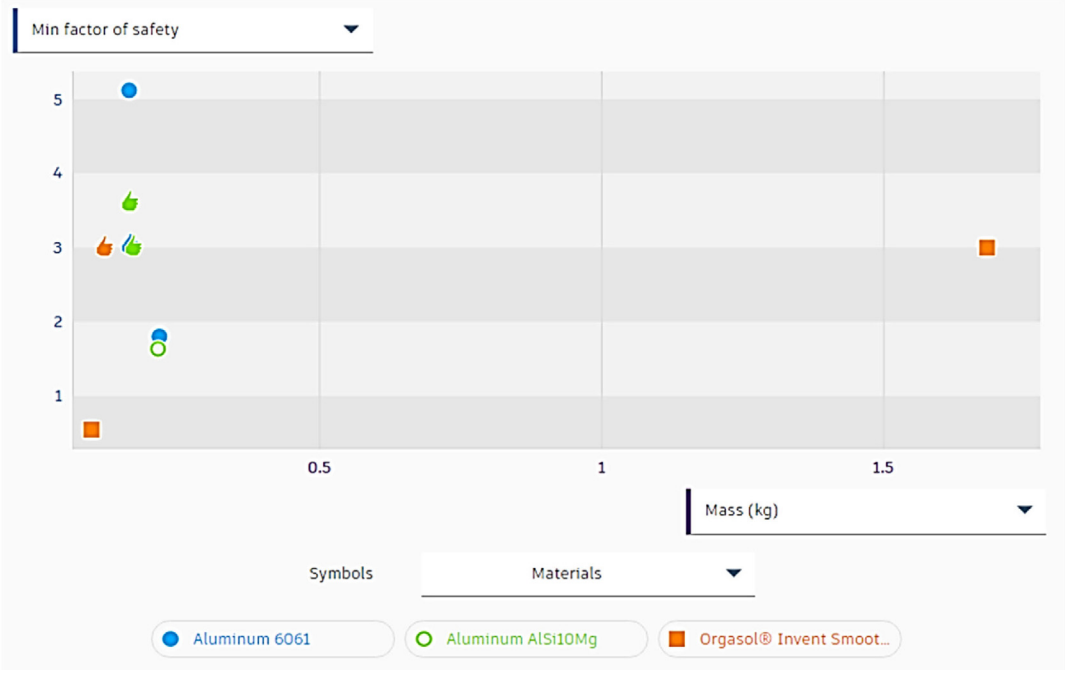


Fig. 5. Comparison plot of minimum FOS and mass.



Fig. 6. Fusion of topology optimization and generative design.

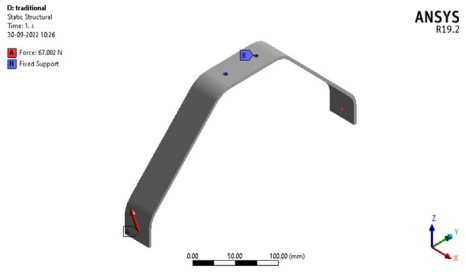


Fig. 7. Boundary condition setup for FEA.

A horizontal and vertical force of 30 N and 60 N have been applied respectively to the axle bearing hole as shown in Fig. 7. Moreover, fixed support had been given to the mounting holes of the landing gear. These boundary conditions have been applied as per the calculations in Section 2.

The horizontal and vertical forces are components of forces acting on the axle bearing hole of the landing gear, which when combined, produces a resultant force of 67.08 N at an angle of 63.43° with respect to the x-axis.

By performing static structural analysis under the mentioned boundary conditions on three models generated as shown in Fig. 8, comparing models based on FOS obtained, the mass of landing gear, and the ratio of FOS and mass have been calculated and presented in Table 4. Tetrahedron elements have been used in the finite element analysis and the mesh for the generative design constitutes 381,086 elements and 591,333 nodes.

In order to choose the best model, it is useful to understand the trade-offs between the model's mass and FOS. The analysis of Table 4 reveals that the generative design landing gear has the

Table 4
FOS by mass comparison table.

Approach	FOS	Mass (grams)	FOS/Mass
Conventional	2.42	215.78	0.0112
Generative	3.12	166.89	0.0187
Fused	2.32	154.8	0.0150

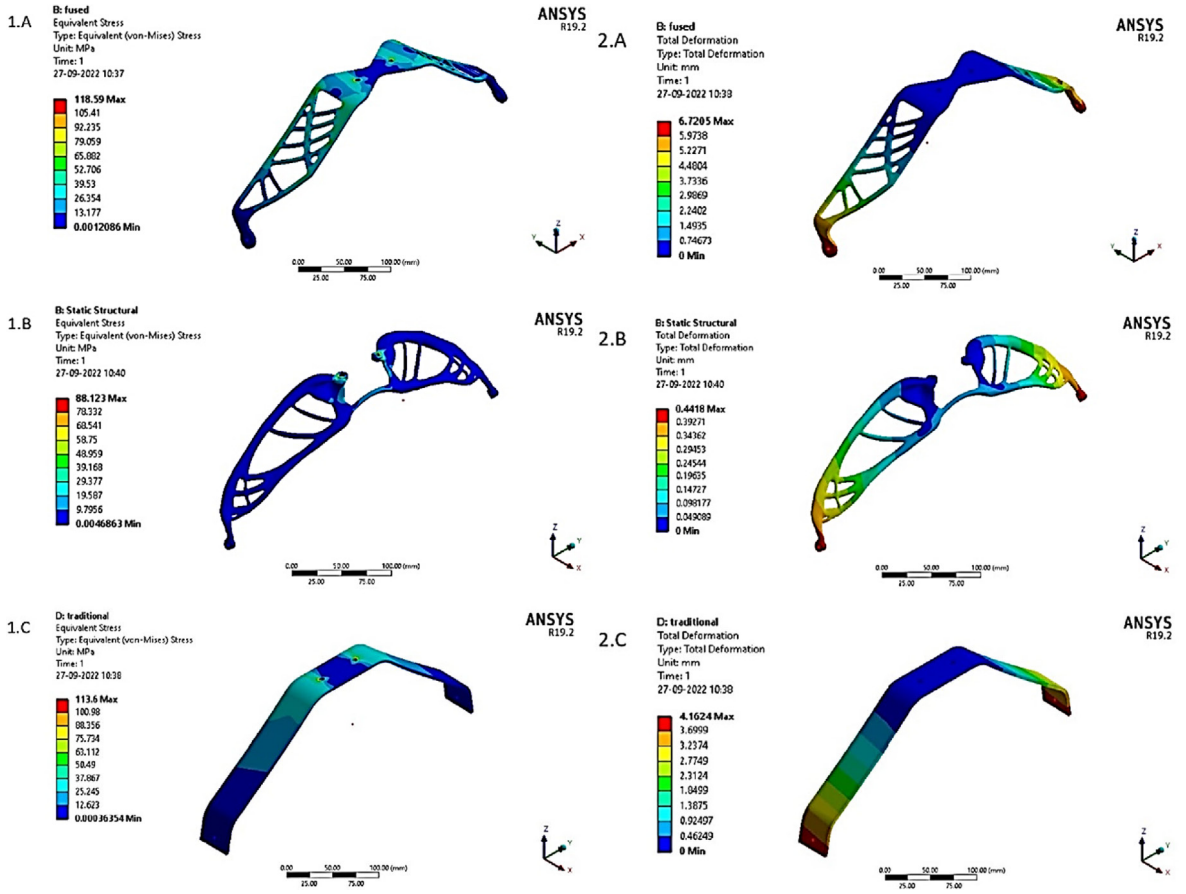


Fig. 8. 1-Equivalent Von-mises stress of landing gears, and 2-Total deformation of landing gears for (A) fusion of topology optimization and generative design (B) generative design (C) conventional design.

highest FOS. The least FOS and mass are evident in the fused model. An aircraft's landing gear is a critical element. It must have a high FOS as well as a low weight to reduce the flying weight. Therefore, the FOS to mass ratio is taken into account when choosing a design. This metric offers an appropriate trade-off between FOS and mass. Hence, based on this metric, the fused design has been chosen as the best design for the landing gear. It is also important to understand the manufacturing expenses of each process. Even though the generative design has the highest FOS and the second lowest mass, the manufacturing cost would be very high, and such expensive components would be difficult to replace if they have been damaged.

5. Conclusions

This paper discusses the objective of an RC plane landing gear and suggests a better design using generative design as a tool. The conventional design has been compared with two different generative designs, which have been differentiated based on the manufacturing method, i.e., additive manufacturing and CNC milling. The results have been validated using static FEA on ANSYS. The final design that had been selected for fabrication resulted in about 30 % weight reduction from the conventional landing gear. The highest FOS and FOS/mass ratio has been achieved in the generative design approach whereas the least FOS and mass have been obtained through a fused design approach with a comparable FOS/-mass ratio. As this generative design technology is in its infant stage, it is important for engineers to pursue the depths of this tool and use it to further show the world the massive potential that it possesses in terms of weight reduction and efficiency.

CRediT authorship contribution statement

Aditya Gupta: Conceptualization, Methodology, Software, Writing – review & editing. **Vedant Soni:** Software, Data curation, Methodology. **Dhaval Shah:** Writing – original draft, Visualization, Investigation, Validation. **Absar Lakdawala:** Supervision, Writing – review & editing.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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